

Supplementary Methods S2 | Online Resource 2

ASSURE-BVLOS: Runtime Assurance for Risk-Adaptive Space–Air–Ground Unmanned Aircraft Operations

Aerospace Systems | Corresponding author: Nick Barua

AN Holdings Co., Nishinomiya, Hyogo, Japan

Shiga University of Medical Science, Otsu, Shiga, Japan

s.nick.barua@gmail.com

Reference implementation: 2.0.0-revision

New computational evaluation; not a recovered original simulation

S2.1 Provenance and limits of the replacement study

The original executable simulation and state-level outputs were unavailable. The previous Version 1.0.0 snapshot supplied documentation and figures only. The revised study newly implements the documented framework, supplies previously missing settings, and replaces all earlier numerical results. It does not establish how the original counts were generated. The newly selected configuration is a synthetic test specification, not a fitted model or an externally preregistered study. Parameters were not adjusted to reproduce the earlier headline result. Table S1 separates retained settings from newly specified values; the executable archive is Online Resource 1 (Supplementary Code S1).

Table S1 Parameter provenance

Retained from the submitted specification	Newly specified for this revision
Five domains; multiplicative complement; separate uncertainty terms	Equal weights 0.20; uncertainty coefficients 0.10; exact normalisation mappings
Authority thresholds 0.25, 0.42, 0.60, 0.76; five response states	Exact baseline, protection/hard cutoffs and fault-flag generators
Three equally likely archetypes; link probabilities 0.35/0.50/0.15	All missing shape, scale, offset and clipping parameters; primary seed 20260918
RTT medians 35/65/560 ms; broad distribution-family choices	Provenance deficit, explicit health deficit and mission-specific age normalisation
Logistic coefficients and interactions; mitigation 0/22/43/66/82%	Paired bootstrap, labels, sensitivity grids, comparators and temporal logic tests

The author-supplied original manuscript is the source for the retained settings. The new implementation and config.json are the source for the revision values. The declared manifest SHA-256 is shown below; it identifies the exact file bytes used in this run, not the correctness or operational suitability of those settings.

82ab6791a44b771a66360d719d94c7ca6a800cd127e0f7e7721d3148fa256baa

S2.2 Complete generator specification

Mission order is disaster response / linear-infrastructure inspection / maritime surveillance. Link order is terrestrial-cellular / LEO-NTN / GEO-SATCOM. Mission probabilities are one-third each; link probabilities are 0.35/0.50/0.15 and are independent of mission. Conditional on these categories, all source variates are drawn independently in the order implemented in generate(). Shared category membership can induce marginal dependence. There is no temporal autocorrelation, latent weather coupling or measurement-error layer.

A lognormal is generated as $\exp(N(\log(\text{median}), \sigma^2))$. A gamma uses shape and scale, not rate. All normal distributions below are clipped after sampling, not sampled from a renormalised truncated-normal density. Beta draws are standard beta distributions before any offset and clipping. These details are necessary to reproduce the exact reference population. Table S2 specifies every source distribution; RTT denotes round-trip time and SD standard deviation.

Table S2 Full source distributions

Variable	Distribution and parameters	Clipping
RTT (ms)	Lognormal: medians 35/65/560; log-SD 0.45/0.50/0.30 by link	[0, 4000] ms
Jitter (ms)	Gamma: shape 2/2/2.5; scale 5/10/20 by link	[0, 500] ms
Packet loss	Beta: $\alpha=1.2/1.3/1.5$; $\beta=35/25/20$ by link	[0, 0.35]
Sensor confidence	Normal: mean 0.75/0.87/0.80; SD 0.10/0.07/0.09 by mission	[0.05, 0.99]
Disagreement	Beta(2,12) + mission offset 0.08/0/0.04	[0,1]
Provenance deficit	Beta(1.5,12) + mission offset 0.04/0/0.06	[0,1]
External-data age (h)	Lognormal: medians 2/4/8; log-SD 0.65/0.70/0.80 by mission	[0,72] h
Weather score	Beta: $\alpha=2.5/2/2.5$; $\beta=3.5/6/4.5$ by mission	Naturally [0,1]
Exposure score	Beta: $\alpha=2/2.5/1.5$; $\beta=3/3.5/6$ by mission	Naturally [0,1]
Energy reserve	Normal: mean 0.60/0.70/0.50; SD 0.16/0.12/0.17 by mission	[0,1]
Health deficit	Beta(2,12)	Naturally [0,1]
Invalid-navigation flag	Bernoulli $p=0.0015/0.001/0.001$ by mission	Boolean
Flight-control-failure flag	Bernoulli $p=0.0005$ in all missions	Boolean
Geofence-breach flag	Bernoulli $p=0.0005/0.001/0.0005$ by mission	Boolean
Paired outcome variate	One independent uniform U per state, reused by both policies	[0,1)

Fault flags are independent logic challenges, not calibrated failure probabilities. They determine required authority floors and critical labels. The surrogate uses the five continuous scores and does not add a fault-flag-specific consequence term. Therefore the simulation cannot estimate real consequences of navigation loss, flight-control failure or geofence breach.

S2.3 Normalisation, authority and outcome algorithms

Table S3 gives the normalisation functions and the two separately applied uncertainty penalties.

Table S3 Normalisation and uncertainty mappings

Quantity	Exact mapping
$c(q)$	$\min(1, \max(0, q))$
r_{C2}	$0.55c(\text{RTT}/800 \text{ ms}) + 0.20c(\text{jitter}/100 \text{ ms}) + 0.25c(\text{loss}/0.20)$
r_s	$1 - \text{confidence}$
r_D	Provenance deficit
r_E	$0.60 \text{ weather} + 0.40 \text{ exposure}$
r_V	$0.65(1 - \text{energy reserve}) + 0.35 \text{ health deficit}$
D_s	Disagreement, excluded from r_s
A_d	$1 - \exp(-\text{age}/\tau)$; $\tau=6/12/18 \text{ h}$ by mission; age excluded from r_D
Weights / penalties	All five $w=0.20$; $\lambda_s=\lambda_a=0.10$

Aggregate the active domains using the multiplicative complement in manuscript Eq. (1); omit zero weights. Numerically, model.py uses log1p and expm1, retaining the limiting value at a unit-valued active domain. Add the two penalties and clip to [0,1]. The band assignment is the count of thresholds less than or equal to the score, so equality enters the higher state. The hard floor is S4 if flight control is flagged failed or $\text{energy} \leq 0.12$; otherwise at least S3 for invalid navigation, geofence breach or $r_{C2} \geq 0.90$. The adaptive policy also requires S2 when $r_{C2} \geq 0.60$. The baseline is precisely the shared hard floor; the adaptive request is the maximum of band, shared hard floor and additional C2 floor.

Manuscript Eqs. (5)–(6) give the complete retained logistic surrogate and paired outcome rule. The state-mitigation vector is $m=[0,0.22,0.43,0.66,0.82]$. Each policy multiplies the unmitigated probability by one minus the mitigation for its assigned state. Generate paired Bernoulli outcomes using one shared uniform variate per sampled state. Under nominal mitigation, an adaptive outcome cannot occur without a baseline outcome: the nesting is mathematically imposed. A mitigation scale is applied to both policies, capped at 0.95. The adverse-intervention variant adds $k \cdot 1(s_A > s_B)$ to p_A and clips at one; additional assignments include S1.

S2.4 Labels, bootstrap and denominators

Compound stress is the union of ($r_{C2} \geq 0.50$ and $r_S \geq 0.25$), ($r_E \geq 0.45$ and $r_V \geq 0.40$), and ($r_D \geq 0.25$ and $r_S \geq 0.25$). Critical states are compound stress or any shared hard override. In the primary population, 10,046 states meet the compound definition and 10,690 meet the critical definition. Recall is the fraction of critical states assigned S2–S4; burden is the fraction of all 100,000 states assigned S2–S4. The noncritical intervention fraction conditions on the noncritical denominator. Labels are not independent truth: they share complete input vectors with the decision rule. During ablation, the labels and hard floors are not recomputed from an incomplete sensor vector.

The realised baseline and adaptive estimates are 10,739/100,000 and 8,083/100,000, respectively. Their paired difference is 0.02656, or 2.656 percentage points. The four empirical joint-cell probabilities correspond to counts [89,261, 0, 2,656, 8,083] ordered as neither, ASSURE only, baseline only and both. Each bootstrap replicate draws Multinomial(N, counts/N), then calculates (baseline-only–ASSURE-only)/N. This is the exact resampling distribution of any count-based paired statistic under an empirical nonparametric bootstrap; it is not an independent-binomial approximation. The relative statistic divides by that replicate's baseline proportion.

Using 10,000 replicates and seed 20261001, percentile limits are [0.02557,0.02757] for the absolute difference and [0.23920338,0.25551523] for the relative difference. Percentages are dimensionless reporting conventions. Neither the bootstrap nor the sampled states provide events per hour or risk per mission. Conditional means before outcome sampling are 0.1059573459 and 0.0796261977. They are intentionally distinct from the realised proportions.

S2.5 Complete analysis inventory

Table S4 lists the analysis grids and their corresponding machine-readable result files.

Table S4 Declared analyses and output files

Analysis	Settings	Output
Threshold sweep	25 multipliers 0.70...1.30, step 0.025	threshold_sweep.csv
Ablation	Five omissions; renormalise; sensing/data omission removes associated penalty; hard channels retained	domain_ablation.csv
Vehicle weight	0...0.60, step 0.10; each other weight=(1- w_V)/4	vehicle_weight_sweep.csv
Random weights	200 Dirichlet(20,20,20,20,20); seed 20261002	random_weight_sensitivity.csv
Uncertainty	λ_s and λ_a each 0,0.05,0.10,0.15,0.20	uncertainty_penalty_sweep.csv
Surrogate	Linear coefficient multipliers 0.8/1/1.2 $\times$ intercepts -5.3/-4.8/-4.3; interactions fixed; plus one no-interaction case	surrogate_sensitivity.csv
Mitigation	Scales 0/0.5/1/1.5, cap 0.95; both policies	mitigation_sensitivity.csv

Analysis	Settings	Output
Adverse intervention	Added conditional $k=0/0.02/0.05/0.10$; bisection break-even	intervention_risk_sensitivity.csv
Aggregation	Multiplicative, arithmetic and maximum; same bands and penalties	aggregation_comparison.csv
Matched budget	Mandatory states retained; stable descending-score ranking to 9,768 selections	matched_budget_comparison.csv
Repeated seeds	20260918...20260927; 100,000 states per seed	seed_results.csv
Temporal logic	500 traces $\times$ 600 ticks; seed 20261003	temporal_logic_results.csv

All sensitivity rows use the identical primary population except the explicitly repeated seeds and temporal traces. The matched-budget diagnostic retains shared hard conditions and moderate-C2 protection as mandatory selections, then fills the remaining budget by score. It is retrospective and should not be interpreted as performance on a held-out calibration/test split. Dirichlet draws are local weight perturbations rather than global Sobol indices. The arithmetic comparator's slightly higher matched-budget recall is retained in the main text. Figure S1 shows the vehicle-weight trade-off.

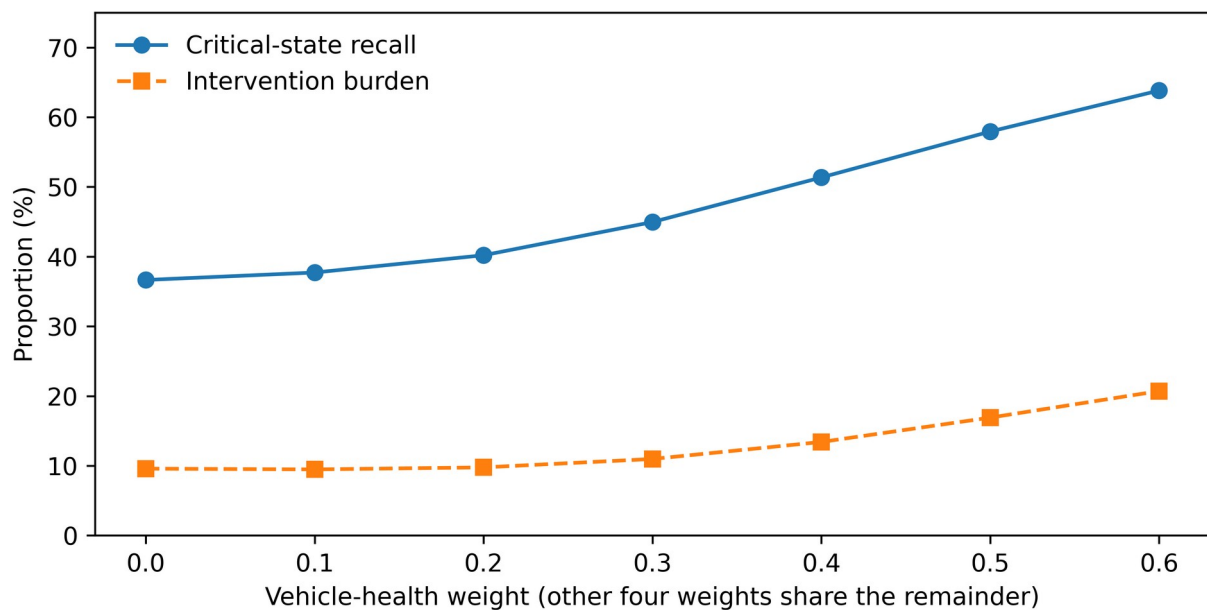


Fig. S1 Vehicle weighting changes recall and burden

Increasing vehicle weight reduces the other four weights because they sum to one. The chart therefore does not demonstrate a free improvement in all domain sensitivities. The nominal point is $w_v=0.20$. Exact estimates and the full weight vectors are saved with the results. Figure S2 then shows the separate adverse-intervention sensitivity.

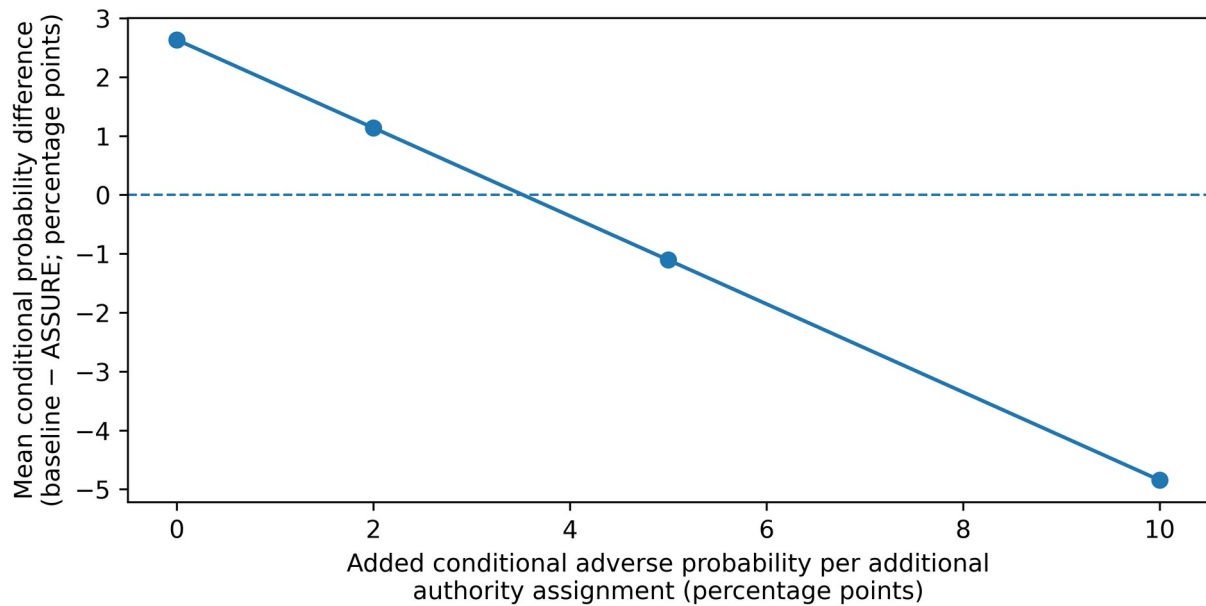


Fig. S2 Sensitivity to an adverse-intervention assumption

The zero crossing is approximately $k=0.0352153$ per additionally assigned state; the affected fraction is 0.74772, including S1. At $k=0.05$ the conditional probability difference is negative. These values are deliberately assumed stress settings, not measurements of the cost or danger of a real intervention. The reversal prevents interpreting positive nominal mitigation as an independently established treatment benefit.

S2.6 Prescribed-score temporal test

Each trace contains six consecutive 100-tick blocks centred at 0.22, 0.43, 0.39, 0.62, 0.37 and 0.20. Independent normal noise of SD 0.02 is added and clipped. Ticks are integer-labelled from 0 to 599, with a nominal 1 s interval. Floor S2 applies to ticks 150–154 inclusive and S3 to 350–352 inclusive. Recovery is allowed in these traces. The supervisor escalates immediately; it lowers by one band only after ten consecutive samples below the current entry threshold minus 0.04. S4 is latched until reset. Separate unit tests cover S4 latching, blocked recovery, invalid scores and override priority, rather than claiming the fixed trace population exercises every exceptional path.

Mean state changes decrease from 97.556 to 5.836, while mean protected-state occupancy increases from 0.2969867 to 0.6372733. The total switching counts are 48,778 and 2,918. There are zero floor violations and zero requested states below the corresponding instantaneous target in 300,000 evaluated ticks. This is not a simulation of a flight, mission delay, pilot workload or physically bounded transition time. Fig. S3 shows the first generated trace, not a selected best trace.

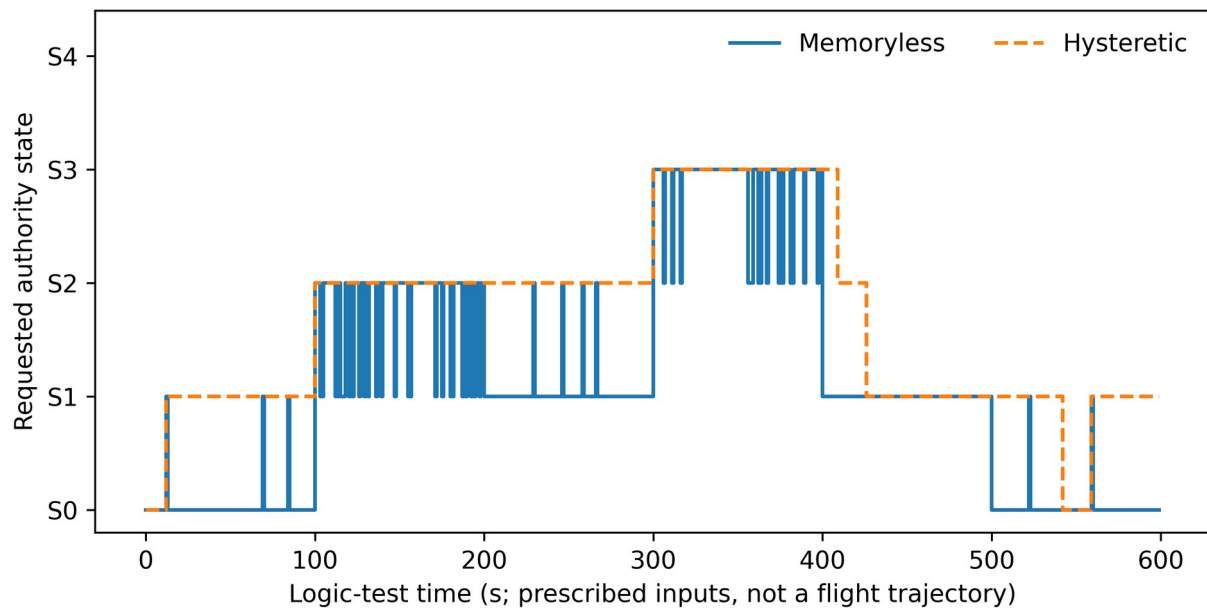


Fig. S3 Example prescribed-score authority trace

S2.7 Reproduction and software checks

The supplied run was executed using Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0 and pandas 2.2.3. Figure generation used Matplotlib 3.10.8. Exact numerical replay was checked in the same environment. Reinstalling the environment on another platform was not independently tested; requirements.txt records the versions used. Small floating-point or random-distribution implementation differences across environments may affect strict array equality, so preserve the tested versions for byte-level checks.

Extract ESM_1.zip and, from its Supplementary_Code_S1 folder, run the following commands.

Reproduction needs no online dataset, external service, graphics processor or operational system after the dependencies are installed. The alternate replay directory preserves the supplied results for comparison.

```
python -m unittest discover -s tests -v
python run_study.py --config config.json --output replay
python verify_outputs.py --results replay
python make_figures.py --results replay --output replay_figures
```

The 28 implementation tests cover scalar bounds, monotonicity, endpoints, zero weights, equal-score identity, arithmetic/maximum inequalities, invalid inputs, threshold equality, hard-override priorities, nested outcomes, zero mitigation and supervisor recovery/override behaviour. This is a finite test suite, not formal software verification or a qualified aviation test campaign. The primary replay regenerates all input arrays and policy outputs and checks exact array equality. verify_outputs.py additionally checks supplied array hashes, re-evaluates primary outcomes and checks consistency of headline and scenario totals.

S2.8 Data dictionary and file roles

primary_states.npz is the complete machine-readable primary population. Each one-dimensional array has N entries; risk has shape $(N,5)$ in C2/Sensing/Data/Environment/Vehicle order. scenario and link are integer category codes 0–2 in the order stated above. Raw observables use the units in Table S2; bounded scores, probabilities and uniform variates are dimensionless. Fault flags, labels and outcomes are Boolean. Authority states are integer codes 0–4. The file also stores the aggregate decision score and both conditional probabilities. data_dictionary.csv describes every saved array. primary_states_sample_1000.csv is only a convenience excerpt, not the analysis population.

All result CSVs are generated outputs rather than hand-entered replacements. `summary.json` contains the primary estimates, intervals, seed summaries and software environment. `replay_checks.json` records exact replay checks; `primary_array_sha256.json` contains array-byte hashes. `unit_tests.txt` records the executed unit-test output. `equations.tex` contains LaTeX sources matching the editable Word equations. The publication's numerical figures are generated from these saved outputs. The three conceptual diagrams are vector redraws of the proposed architecture, authority bands and claim hierarchy; they are not numerical simulation outputs. Their artwork is generated separately by `draw_conceptual_figures.py` using ReportLab

ReportLab and Poppler are optional artwork tools rather than simulation dependencies. The revised reproducibility package, including Supplementary Code S1 and Supplementary Methods S2, is publicly archived on Zenodo as Version 2.0.0-revision (DOI: 10.5281/zenodo.22827977).